🧑‍🏫 **Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 5 · STANDARD 6.GR.1

Area of Regular Polygons

Lesson 5-9 · Teacher Copy

Name: Type your name **Date:** Today's date **Class:** Class period

My Notes Level 3 Enrichment

🎯 **TODAY'S OBJECTIVES**

Content Objective: I can find the area of a regular polygon by decomposing it into triangles.
Language Objective: I can explain my steps using the words regular polygon, decompose, triangle, and composite.

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

Area of Regular Polygons | **Regular Polygon** | **Decompose** | **Triangle** | **Composite** | **Formula**

👆 *Tap any word to see what it means and a picture.*

Level 3: try the blanks from memory first, then check the bank.

1. The space inside a regular polygon, found with $A = (\frac{1}{2}bh) \times n$, is the ________.

2. A polygon with all sides and all angles equal is a ________.

3. To break a shape into smaller shapes is to ________ it.

4. A three-sided polygon is a ________.

5. Made of two or more shapes combined, a figure is ________.

6. A math rule written with symbols is a ________.

Write About the Math

How much will the gazebo flooring cost?
¿Cuánto costará el piso del quiosco?

Say your answer to a partner first. Then write it, using at least two of these words:

- [ ] **Area of Regular Polygons** — *Área de polígonos regulares*
- [ ] **Regular Polygon** — *Polígono regular*
- [ ] **Decompose** — *Descomponer*
- [ ] **Triangle** — *Triángulo*
- [ ] **Composite** — *Compuesto*

Model: *We decompose the hexagon into triangles because we already know how to find a triangle's area.* — Students explain that splitting the hexagon into 6 equal triangles lets them use the triangle area formula they already know, then combine the parts.

Start

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **One triangle's area is ____ sq ft. The octagon has ____ triangles, so total area = ____ sq ft. The cost is ____ x $3 = $____.**
- **My answer is ____ because ____.**
  Mi respuesta es ____ porque ____.

Build

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ____.**
  Mi afirmación es ____.
- **My evidence is ____, so ____.**
  Mi evidencia es ____, entonces ____.

Explain

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- [ ] I answered the question.
- [ ] I used math evidence: numbers, an equation, or a model.
- [ ] I used at least two math words.

Writing Guide — Teacher Copy

- The answer to the math question is correct.
- The evidence is real lesson mathematics (numbers, equation, or model), not restated claim.
- The support level changed the language scaffolding, never the mathematical expectation.

Answer Key & Teacher Guide

1. **Notes 1:** Area of Regular Polygons
2. **Notes 2:** Regular Polygon
3. **Notes 3:** Decompose
4. **Notes 4:** Triangle
5. **Notes 5:** Composite
6. **Notes 6:** Formula
7. **Watch for this mistake:** *A common mistake in Area of Regular Polygons is stopping after finding the area of ONE triangle and reporting that as the total area — or adding the number of triangles instead of multiplying. Remember: total area = (one triangle's area) × (number of sides), not (one triangle's area) + (number of sides).*
8. **Try It 1:** A. 60 in² — *One triangle = ½ × 4 × 6 = 12 in². A pentagon has 5 triangles, so total = 5 × 12 = 60 in².*
9. **Try It 2:** A. 120 sq ft — *An octagon has 8 sides, so it splits into 8 triangles. One triangle = ½ × 6 × 5 = 15 sq ft. Total = 8 × 15 = 120 sq ft.*